Trainings

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the engine oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)
- Remove the pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)

- Remove the suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)
- Remove the transfer tube. Refer to Procedure 007-040 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-040-tr.html)
- Remove the oil pump. Refer to Procedure 007-031 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)

Note : Record the quantity and location of the shims under the oil pump feet. The same number of shims must be put back in the same location when the oil pump is installed.

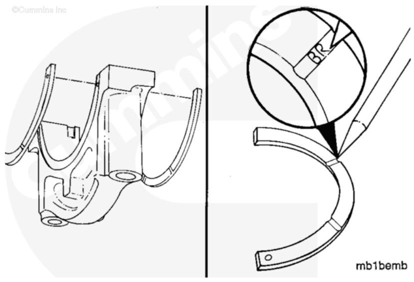
- Remove the stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-089-tr.html)

Remove

The number 6 main bearing cap on QSK45 engines and the number 8 main beraring cap on QSK60 engines contain two thrust bearings, each on the front and rear of the cap.

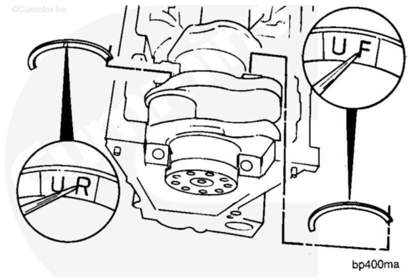
The main bearing cap **must** be removed for the thrust bearing to be removed.

Remove the two thrust bearings from the bearing cap, and mark them for position in the notched area.



Note : If necessary, slide the crankshaft to the front or to the rear to allow the thrust bearings to be removed.

Remove the two upper thrust bearings from the block at the appropriate location. Mark the thrust bearings for position.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

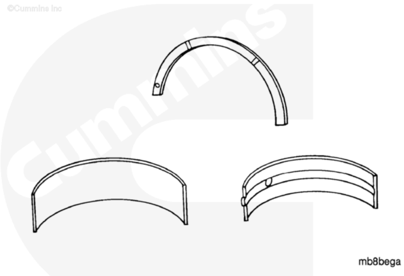
Wear appropriate eye and face protection when using compresed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

To reduce the possibility of bearing damage, do not use a scraper or wire brush to clean bearings.

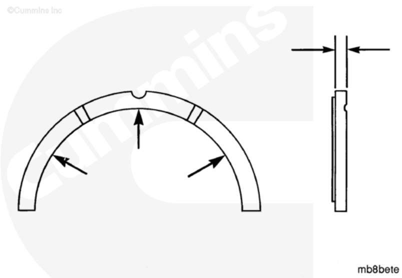
Clean the bearings with solvent and dry with compressed air.

Make sure the bearings are marked for location. The bearing **must** be installed in the original location if used again.



Use a micrometer to measure the bearing thickness at the wear location.

The areas that shine indicate the location where the crankshaft contacts the bearing.



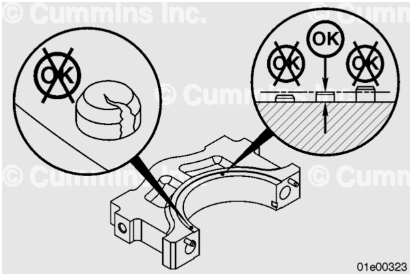
Thrust Bearing Thickness

	mm		in
Standard	7.239	MIN	0.285
	7.315	MAX	0.288
Oversize 0.254 mm [0.010 in]	7.49	MIN	0.295
	7.57	MAX	0.298
Oversize 0.508 mm [0.020 in]	7.75	MIN	0.305
	7.82	MAX	0.308
Oversize 0.762 mm [0.030 in]	8.00	MIN	0.315

	8.08	MAX	0.318
Oversized 1.016 mm [0.040 in]	8.25	MIN	0.325
	8.33	MAX	0.328

Check the thrust bearing retaining pin in the main bearing cap for mechanical damage.

Check the thrust bearing retaining pins protrusion.



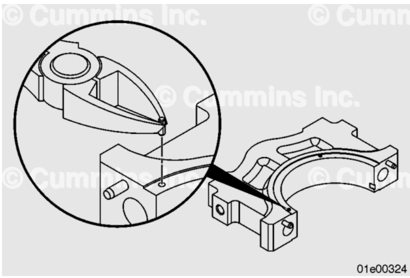
Dowel Pin Protrusion		
mm		in
4.0	MIN	0.157
5.0	MAX	0.197

If the thrust bearing retaining pins are damaged or having incorrect protrusion, they **must** be replaced.

Repair

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Note : If one or more pins are damaged, replace all four pins.

Use a pair of cutting dikes and grasp the damaged pin.

Place a piece of thin wood, or equivalent, between the dikes and main bearing cap, if necessary.

Pry the damaged pin out of the bearing cap.

Use compressed air to blow any dirt or material out of the pin hole.

The pin insertion tool can be fabricated locally.

The tool is to be made from 4340 steel, or equivalent.

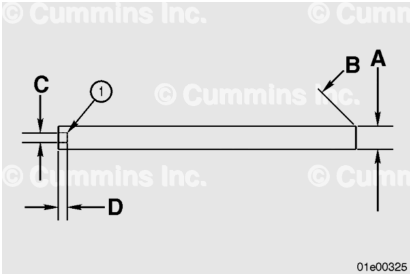
The material will need to be thoroughly hardened to 45-50 HRC.

Fabricate the pin insertion tool using the following dimensions.

The overall length of the tool is 160 mm [6.299 in].

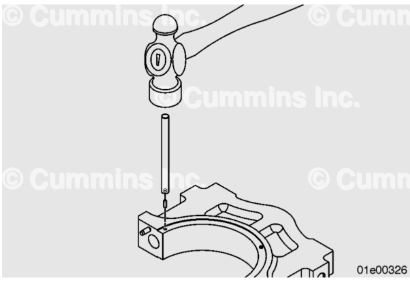
A. 12.5 mm [0.492 in]
B. 0.5 mm [0.020 in]
C. 5.2 mm [0.205 in]
D. 5 mm [0.197 in].

1. Chamfer should be no greater that 0.4 x 0.2 mm [0.016 x 0.008 in].



⚠ WARNING ⚠

To reduce the possibility of personal injury or equipment damage, this procedure must only be performed by suitably qualified service technicians.



Install the new thrust bearing retaining pins (two on each side) into the main bearing cap to facilitate the installing of the lower thrust bearings.

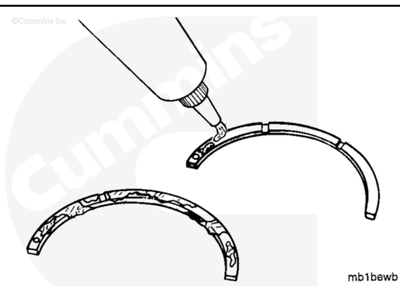
To avoid thrust bearing retaining pin damage, use the pin insertion tool and a copper hammer.

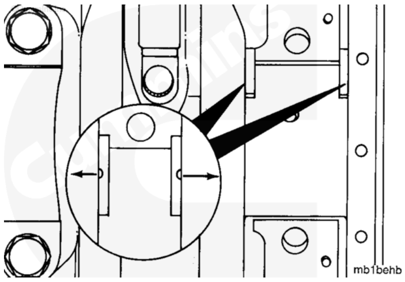
This tool will set the pins to the correct depth of protrusion.

Check the thrust bearing retaining pin protrusion after installation. Reference the Dowel Pin Protrusion information in the Inspect for Reuse section of this procedure.

Install

Lubricate the grooved side of the thrust bearings with Lubriplate™ 105, Part Number 3163086.





⚠ CAUTION ⚠

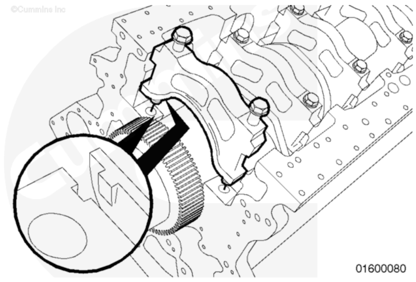
The grooves in the thrust bearings must point toward the crankshaft.

Note : Move the crankshaft to the front or rear so the second thrust bearing can be installed.

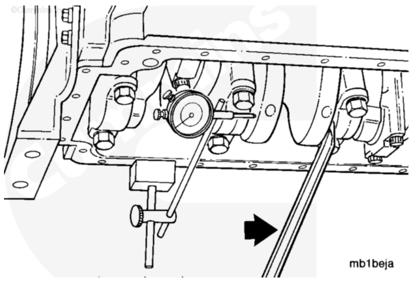
Note : The upper and lower thrust bearing on the same side of the main bearing cap must be the same thickness.

Install the upper thrust bearings in the number 6 main bearing cap for the QSK45 and the number 8 main bearing cap for QSK60 engines.

Slide the lower thrust bearings over the pins into the machined lip of the main bearing caps.



Install the main bearing cap. Refer to Procedure 001-006 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-006.html)



Use a dial indicator, Part Number 3376050, and magnetic base, Part Number 3377399, to measure the crankshaft's end clearance.

Crankshaft End Clearance		
mm		in
0.13	MIN	0.005
0.51	MAX	0.020

If the clearance is **not** within specifications, check for foreign material behind the thrust bearings. Oversize thrust bearings are available to adjust the end clearance.

Note : The upper and lower thrust bearing on the same side of the main bearing cap must be the same thickness.

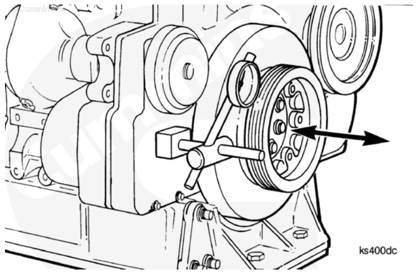
Note : It is recommended to recheck the crankshaft end clearance once the engine is installed into an

application.

Release the tension on the cooling fan drive belt, if installed. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-002-tr.html)

Remove an accessible handhole cover. Refer to Procedure 001-035 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-035-tr.html)

Use the barring device to rotate the engine so that the connecting rod journal is opposite the handhole cover opening. If the connecting rods are already in this position when the handhole cover is removed, use the barring device to rotate the crankshaft slightly in the opposite direction to normal rotation. This releases the loading on the gear train.



Use a pry bar or similar to move the crankshaft backward by prying against the crankshaft web.

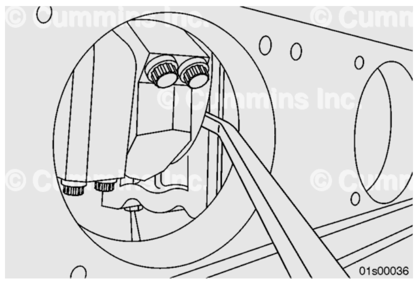
Note : Do not pry against the connecting rods or the crankshaft counterweights.

Set the dial indicator to zero.

Move the crankshaft in the opposite direction by prying on the crankshaft web.

Check the reading on the dial indicator to confirm the crankshaft has end clearance that is within the limits shown above.

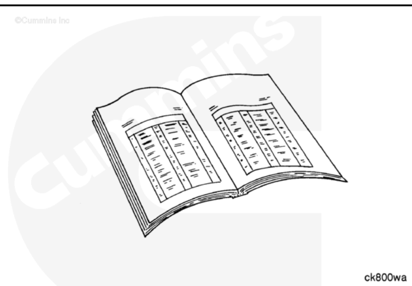
If previously loosened, tension the cooling fan drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-002-tr.html)



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-089-tr.html)
- Install the oil transfer tube. Refer to Procedure 007-040 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-040-tr.html)
- Install the oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)
- Install the lube oil pump, if removed. Refer to Procedure 007-031 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)
- Install the oil pan adapter and related components. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)
- Install the oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)
- Fill the engine with clean 15W-40 oil. Refer to Procedure 018-017 in Section V. (/qs3/pubsys2/xml/en/procedures/56/56-018-017.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to 70°C [158°F] coolant temperature and check for leaks.

Last Modified: 01-Jun-2017